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Liu, Dehai (PRC-BJN-MCS); **Wang, Kaishun** (PRC-BJN-MCS);
Yao, Tian (PRC-HENST-SM)

***s*-almost cross-*t*-intersecting families for finite sets.** (English. English summary)

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A family $\mathcal{F} \subseteq \binom{[n]}{k}$ is called *t-intersecting* if $|F \cap F'| \geq t$ for all $F, F' \in \mathcal{F}$, and two sets F, G are said to be *t-disjoint* if $|F \cap G| < t$. The structure of extremal *t*-intersecting families was completely determined by the theorem of P. Erdős, C. Ko and R. Rado [Quart. J. Math. Oxford Ser. (2) **12** (1961), 313–320; MR0140419] and the complete intersection theorem of R. F. Ahlswede and L. H. Khachatrian [European J. Combin. **18** (1997), no. 2, 125–136; MR1429238]. Two families $\mathcal{F}, \mathcal{G} \subseteq \binom{[n]}{k}$ are said to be *cross-t-intersecting* if $|F \cap G| \geq t$ for all $F \in \mathcal{F}$ and $G \in \mathcal{G}$, and there is a vast literature on determining the maximum product $|\mathcal{F}||\mathcal{G}|$ of sizes of such families; see N. Tokushige [J. Algebraic Combin. **38** (2013), no. 3, 653–662; MR3104733] for an eigenvalue method approach, and H. Zhang and B. Wu [J. Combin. Theory Ser. B **171** (2025), 49–70; MR4836581] for the resolution of a conjecture of Tokushige.

The authors consider the following relaxation: $\mathcal{F}$ and $\mathcal{G}$ are called *s-almost cross-t-intersecting* if each member of $\mathcal{F}$ (resp. $\mathcal{G}$) is *t*-disjoint with at most *s* members of $\mathcal{G}$ (resp. $\mathcal{F}$). The single-family analogue (i.e., *s*-almost *t*-intersecting families) has been studied by D. Gerbner et al. [SIAM J. Discrete Math. **26** (2012), no. 4, 1657–1669; MR3022158] and P. Frankl and A. B. Kupavskii [Electron. J. Combin. **28** (2021), no. 2, Paper No. 2.7; MR4245300], among others. For the cross version, Gerbner et al. [Graphs Combin. **29** (2013), no. 3, 489–498; MR3053596] determined the maximum value of $|\mathcal{F}||\mathcal{G}|$ in the case $t = 1$.

In this paper, the authors find the maximum product $|\mathcal{F}||\mathcal{G}|$ for all $t \geq 1$. Write $\mathcal{H}_1([n], W; k) = \{H \in \binom{[n]}{k} : W \subseteq H\}$ for the full *t*-star centered at a *t*-set *W*. The authors show that for $k \geq t + 1$ and $n \geq (t + 1)(2(k - t + 1)^2 + 7s)$, the maximum is attained if and only if $\mathcal{F} = \mathcal{G} = \mathcal{H}_1([n], W; k)$ for some $W \in \binom{[n]}{t}$. Note that this is the same extremal configuration as in the cross-*t*-intersecting case; the almost-intersection parameter *s* does not enlarge the maximum product.

The authors also characterize the extremal families among *s*-almost cross-*t*-intersecting families that are not cross-*t*-intersecting. For $k \geq t + 2$, $|\mathcal{F}| \leq |\mathcal{G}|$, and *n* sufficiently large, they show that there exist $X \in \binom{[n]}{k+1}$ and $W \in \binom{X}{t}$ such that $\mathcal{F} = \mathcal{H}_1([n], W; k) \setminus \mathcal{A}$ and $\mathcal{G} = \mathcal{H}_1([n], W; k) \cup \mathcal{B}$, where $\mathcal{A}$ is a $(\binom{n-k-1}{k-t} - s)$ -subset of $\{F \in \binom{[n]}{k} : F \cap X = W\}$ and $\mathcal{B}$ is a $\min\{t, s\}$ -subset of $\binom{X}{k} \setminus \mathcal{H}_1(X, W; k)$. The case $k = t + 1$ is a bit more delicate: when $t \geq s + 2$, one family is a singleton $\{Y\}$ and the other is $\{G \in \binom{[n]}{t+1} : |G \cap Y| \geq t\}$ together with *s* additional sets; when $t \leq s + 1$, the families are built around a $(t + s + 2)$ -set with a balanced covering property. By combining these results with a stability result for cross-*t*-intersecting families due to M. Cao et al. [European J. Combin. **120** (2024), Paper No. 103958; MR4728106], the authors obtain a complete characterization of the extremal *s*-almost cross-*t*-intersecting families subject to $|\bigcap_{H \in \mathcal{F} \cup \mathcal{G}} H| < t$; the answer depends on whether $k \geq 2t + 1$ or $k \leq 2t$ (with the exception $(k, t) = (4, 2)$).

The proofs are organized around case analysis on the pair of *t*-covering numbers $(\tau_t(\mathcal{F}), \tau_t(\mathcal{G}))$, where $\tau_t(\mathcal{F})$ denotes the minimum size of a subset $T \subseteq [n]$ satisfying $|T \cap F| \geq t$ for all $F \in \mathcal{F}$. A Bollobás-type set-pair inequality due to S. Oum and S. Wee [Combinatorica **38** (2018), no. 5, 1279–1284; MR3884790] is used to handle the cases where the covering numbers exceed *k*. The authors remark that the methods can also be applied to subspaces of a vector space over a finite field. *Brahadeesh Sankarnarayanan*

[References]

1. R. Ahlswede, L.H. Khachatrian, The complete nontrivial-intersection theorem for systems of finite sets, *J. Comb. Theory, Ser. A* 76 (1996) 121–138. MR1405994
2. R. Ahlswede, L.H. Khachatrian, The complete intersection theorem for systems of finite sets, *Eur. J. Comb.* 18 (1997) 125–136. MR1429238
3. M. Alishahi, A. Taherkhani, Extremal G -free induced subgraphs of Kneser graphs, *J. Comb. Theory, Ser. A* 159 (2018) 269–282. MR3820368
4. P. Borg, The maximum product of weights of cross-intersecting families, *J. Lond. Math. Soc.* 94 (2016) 993–1018. MR3614937
5. M. Cao, M. Lu, B. Lv, K. Wang, Nearly extremal non-trivial cross- t -intersecting families and r -wise t -intersecting families, *Eur. J. Comb.* 120 (2024) 103958. MR4728106
6. P. Erdős, C. Ko, R. Rado, Intersection theorems for systems of finite sets, *Q. J. Math. Oxf. Ser. (2)* 12 (1961) 313–320. MR0140419
7. P. Frankl, The Erdős-Ko-Rado theorem is true for $n = ckt$, in: *Combinatorics*, vol. I, Proc. Fifth Hungarian Colloq., Keszthely, 1976, in: *Colloq. Math. Soc. János Bolyai*, vol. 18, North-Holland, 1978, pp. 365–375. MR0519277
8. P. Frankl, On intersecting families of finite sets, *J. Comb. Theory, Ser. A* 24 (1978) 146–161. MR0480051
9. P. Frankl, Z. Füredi, Non-trivial intersecting families, *J. Comb. Theory, Ser. A* 41 (1986) 150–153. MR0826944
10. P. Frankl, A. Kupavskii, Almost intersecting families, *Electron. J. Comb.* 28 (2021), #P2.7. MR4245300
11. P. Frankl, J. Wang, A product version of the Hilton-Milner-Frankl theorem, *Sci. China, Math.* 67 (2024) 455–474. MR4698775
12. D. Gerbner, N. Lemons, C. Palmer, B. Patkós, V. Szécsi, Almost intersecting families of sets, *SIAM J. Discrete Math.* 26 (2012) 1657–1699. MR3022158
13. D. Gerbner, N. Lemons, C. Palmer, D. Pálvölgyi, B. Patkós, V. Szécsi, Almost cross-intersecting and almost cross-sperner pairs of families of sets, *Graphs Comb.* 29 (2013) 489–498. MR3053596
14. A. Hilton, E. Milner, Some intersection theorems for systems of finite sets, *Q. J. Math. Oxf. Ser. (2)* 18 (1967) 369–384. MR0219428
15. L. Ji, D. Liu, K. Wang, T. Yao, S. Yu, s -almost t -intersecting families for vector spaces, *Discrete Math.* 349 (2026) 114661. MR4929523
16. D. Liu, K. Wang, T. Yao, s -almost t -intersecting families for finite sets, arXiv:2410.20185.
17. D. Liu, J. Wang, T. Yao, s -almost cross- t -intersecting families for vector spaces, in preparation. MR4492121
18. M. Matsumoto, N. Tokushige, The exact bound in the Erdős-Ko-Rado theorem for cross-intersecting families, *J. Comb. Theory, Ser. A* 52 (1989) 90–97. MR1008163
19. S.-I. Oum, S. Wee, A remark on the paper “properties of intersecting families of ordered sets” by O. Einstein, *Combinatorica* 38 (2018) 1279–1284. MR3884790
20. L. Pyber, A new generalization of the Erdős-Ko-Rado theorem, *J. Comb. Theory, Ser. A* 43 (1986) 85–90. MR0859299
21. A. Scott, E. Wilmer, Hypergraphs of bounded disjointness, *SIAM J. Discrete Math.* 28 (2014) 372–384. MR3176318
22. Y. Shan, J. Zhou, Almost intersecting families for vector spaces, *Graphs Comb.* 40 (2024) 62. MR4742896
23. N. Tokushige, The eigenvalue method for cross t -intersecting families, *J. Algebr. Comb.* 38 (2013) 653–662. MR3104733
24. R.M. Wilson, The exact bound in the Erdős-Ko-Rado theorem, *Combinatorica* 4

(1984) 247–257. MR0771733

25. H. Zhang, B. Wu, On a conjecture of Tokushige for cross- t -intersecting families, J. Comb. Theory, Ser. B 171 (2025) 49–70. MR4836581

Note: This list reflects references listed in the original paper as accurately as possible with no attempt to correct errors.